

MetaCourt: Leveraging the Metaverse for Immersive Court Interpreting Education with Self-Determination Theory

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Figure 1: An overview of MetaCourt and various role and their point-of-view: (1) Judge; (2) Court Clerk, i.e., Interpreter; (3) Jury; (4) Witness; (5) Exhibit.

ABSTRACT

Court interpreting is vital for justice, yet traditional training methods often lack hands-on engagement and rely on uninspiring trial-and-error techniques. To address this, we developed MetaCourt, an immersive virtual environment for court interpreting education. Through participatory design, we created a system evaluated by 21 undergraduate interpreting students using the PENS model based on self-determination theory. Preliminary results showed high autonomy, intuitive controls, and presence, with improved fluency in VR compared to traditional methods. Surveys (NASA-TLX, IPQ) indicated reduced mental workload and strong usability. This study pioneers integrating VR with PENS for court interpreting, offering significant advantages in training effectiveness.

Index Terms: Metaverse, Virtual Reality (VR), Immersive Learning, Self-Determination Theory (STD), Player Experience of Needs Satisfaction (PENS)

1 INTRODUCTION

Court interpreting plays a vital role in the justice system and requires simultaneous and consecutive interpreting skills. While formal instruction provides foundational knowledge, practical courtroom experience is critical for proficiency. However, opportunities

for hands-on training are scarce, forcing novice interpreters to rely on trial-and-error learning in real cases, which can jeopardize justice due to potential misinterpretation. This gap underscores the need for innovative training approaches [6].

Advancements in educational technology, particularly virtual reality (VR), have demonstrated potential in interpreter training [4], yet applications in legal contexts remain underexplored [8]. Most court interpreting research emphasizes the listening and translation approach [1], neglecting the enriched learning environments that VR can offer [2]. Moreover, while Self-Determination Theory (SDT) [7] has been applied to examine virtual environments, its integration with court interpreting education is unprecedented. This study bridges these gaps by leveraging SDT to explore how VR environments satisfy psychological needs, fostering long-term engagement and skill development.

We developed MetaCourt, a VR-based training system that replicates courtroom scenarios using participatory design insights. Users, represented by cartoon-like avatars, engage in interpreting tasks during simulated criminal trials, acquiring hands-on experience. Our research evaluates MetaCourt's impact on interpreting education through four dimensions—General Scoring Technique, PENS, NASA-TLX, and IPQ—comparing traditional PC-based methods to immersive VR environments. The preliminary participant test shows cues on improved fluency, autonomy, reduced workload, and increased presence in VR settings, highlighting the potential of MetaCourt as an efficient alternative to court interpreting education.

2 SYSTEM DESIGN AND IMPLEMENTATION

MetaCourt Development and Participatory Design MetaCourt is a VR environment designed for court interpreting training, providing immersive simulations of courtroom settings. Using participatory design principles, participants contributed feedback on necessary components, user interfaces, and learning content. This two-stage process refined MetaCourt to meet the specific needs of interpreting learners. MetaCourt recreates courtroom scenarios covering criminal procedure and roles such as the judge,

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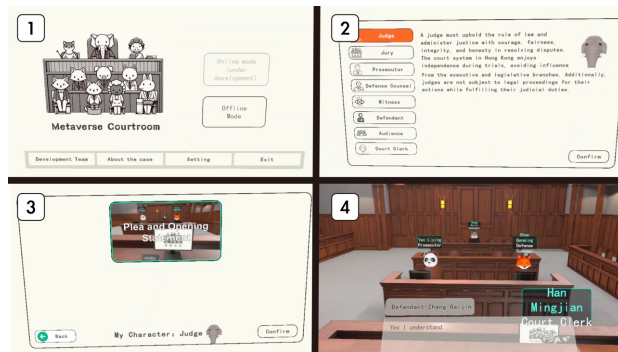


Figure 2: The starting procedures: (1) Choose Offline Mode (2) Choose Judge as Character (3) Choose Scene Plea and Sentence (4) Enter MetaCourt's virtual space.

lawyers, witnesses, and interpreters. Participants engage with different perspectives, enhancing their understanding of real-world legal procedures. In a pre-experiment lecture, participants were introduced to the structure and roles of a Hong Kong criminal court. Key courtroom procedures, such as the prosecution opening and cross-examination, were explained. A glossary of legal terms ensured familiarity with specialized vocabulary. Participants also received VR headset training.

Experimental Procedure A preliminary user study included 21 participants, who acted as court interpreters in MetaCourt, interpreting Chinese questions into English and vice versa during pre-recorded court proceedings. No time limits were imposed, replicating real-time demands. Participants performed the same tasks in a computer lab for the control experiment. The experiment has been granted approval from the Institutional Review Board of the Hong Kong Polytechnic University on 1st April 2024 with project title "Moot Court in the Metaverse - Enhanced Learning Experience with VR", the reference number is HSEARS20240109002.

Participants were split into two groups: one completed the VR test first, while the other began with the control test. Identical pre-recorded proceedings ensured consistency across conditions. MetaCourt's performance was evaluated using fluency, accuracy, cognitive workload (NASA-TLX), and presence (IPQ).

3 RESULTS AND IMPLICATIONS

This study utilized five measures to evaluate the effectiveness of MetaCourt in court interpreting education: interpreting performance (accuracy and fluency), PENS questionnaire (psychological needs), NASA-TLX (cognitive workload), IPQ (sense of presence), and semi-structured interviews.

Accuracy and Fluency in Interpreting Performance Fluency significantly improved in the VR environment compared to PC, while no statistical difference was found for accuracy. This suggests VR's immersive feedback cues and reduced cognitive load, as indicated by NASA-TLX scores, contribute to smoother performance. However, accuracy improvements are likely to require the development of foundational language skills, which is less affected by short-term training [3].

Psychological Needs Satisfaction (PENS), NASA-TLX & IPQ PENS results revealed VR significantly enhanced autonomy, indicating high learner control. However, PC performed better in competence, likely due to participants' familiarity with traditional methods. Relatedness scores were moderate, suggesting potential for stronger social presence. NASA-TLX results indicated that the VR environment reduced cognitive workload, while IPQ findings

highlighted moderate levels of presence, further supporting MetaCourt's usability and immersive design. Correlation analysis highlighted strong links between autonomy, presence, and intuitive controls, suggesting that improving one area could enhance overall immersion and user experience.

Semi-Structured Interviews Participants expressed high satisfaction with MetaCourt, praising its immersive and supportive environment. They found it effective in understanding court interpreting procedures but less effective in improving interpreting skills. Suggestions included incorporating diverse accents and scenarios to better reflect real-world challenges. The comments emphasized the suitability of the system for beginners, with features such as animal avatars that foster a relaxed atmosphere and reduce anxiety.

These findings suggest MetaCourt effectively enhances procedural understanding and fluency, offering promise as a VR-based training tool while highlighting areas for further refinement.

4 CONCLUSION

This study highlights the effectiveness of MetaCourt, a virtual reality system, in improving court interpreting education through realistic and immersive simulations. By promoting autonomy, competence, and relatedness while maintaining cognitive manageability, MetaCourt addresses the limitations of traditional training methods. Its success suggests potential applications in other high-pressure fields, such as medical or diplomatic interpreting, where precision and contextual understanding are critical [5]. This study, while promising, is limited by its small sample size and novice participants, affecting generalizability. Future research should include larger, diverse groups and collaborative designs to enhance immersion. MetaCourt offers a modern, efficient approach to court interpreting education, aligning with global justice standards.

ACKNOWLEDGEMENTS

This research is supported by the Hong Kong Polytechnic University's Start-up Fund for New Recruits (Project ID: P0046056) and a collaborative research project (Project ID: P0047685).

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